LESSON 08 · MISSION GUIDE

A Thousand Coins at Once

NumPy: do the maths on 200 salaries in one line, no loop, no dragging.

From **Orrin of the Clockworks**
To **Jen, Keeper of the Ledger**
By **raven (the third one this week about the loop)**

Keeper. I watched the Grimoire do the pay review at the last full moon. It walks the Roll one subject at a time, recalculating the whole workbook between each. Forty seconds. Sometimes it skips the first row; nobody knows why. Mistress Pell brews tea while it runs. She has a dedicated mug.

NumPy treats a whole column as one thing: `salaries / 12` divides all 200 at once. A legion, not a queue. It's dragging a formula down, except it's instant and never skips the first row. Install it with pip, then go and retire Mistress Pell's mug. — O.

What you'll be able to do after today

- [ ] Turn a list of numbers into a NumPy **array** and ask it for min, max, mean, median, sum
- [ ] Do **column maths** with no loop: `salaries / 52 / hours` gives 200 hourly rates at once
- [ ] Filter with **boolean masks**: `salaries[salaries > 40000]` is AutoFilter in one line
- [ ] Combine filters with `&`, `|` and `~`
- [ ] Compute **percentiles and quartiles**, the ones the Accord of Equal Coin asks for
- [ ] Work with a 2-D array (a whole range) and sum it along either axis

Why this matters at work

- **This is the real pay gap calculation.** Exercise 4 computes Thornmere's mean and median gender pay gap from all 200 subjects. Exercise 5 does the quartiles. Those are two of the six figures the Accord (and the UK regulations) ask for.
- **pandas is built on NumPy.** Next lesson's DataFrame columns *are* arrays. Everything you learn today (masks, axis, vectorised maths) carries over directly.

Four big ideas

1. An array is a column	`np.array([1, 2, 3])`. One type for the whole column (`dtype`), one shape. Like a column of numbers in Excel, not a list of mixed things.
2. Maths is element-wise	`salaries * 1.03` raises every salary. `salaries / hours` divides row by row. No loop. That's "vectorised".
3. A comparison gives a mask	`salaries > 40000` is an array of True/False. Put it in square brackets to filter: `salaries[mask]`. `mask.sum()` counts the Trues.
4. Axis	In a 2-D array, `axis=0` works down the columns, `axis=1` along the rows. SUM at the bottom of a column vs SUM at the end of a row.

Three things that will bite you

- `and` / `or` don't work on arrays. Use `&` / `|`, and put brackets round each comparison.
- `np.percentile(x, 75)`, not `0.75`. Excel's `PERCENTILE` wants 0.75; NumPy wants 75.
- `salaries.mean` without brackets is the method, not the number. If you see `<built-in method mean>`, add `()`.

How today works

15 MIN
Lecture
Lars does the pay review in one line, then explains it

60 MIN
Exercise sheet
Seven ravens about the 200-row Roll of the Realm

15 MIN
Pages from the Grimoire, page 8
Dalvric's quartile report. The Queen still has the error

The ravens

#	FROM	RAVEN	YOU'LL USE
1	Pell	Your first column	`np.array()` `.shape` `.min()` `.mean()` `np.median()`
2	Pell	Dragging a formula down	array maths, `.round(2)`
3	Siv	AutoFilter, but faster	masks, `&` `|` `~`, `.sum()`
4	The Queen	The gap, properly this time	masks by gender, mean vs median
5	The Queen	Pay quartiles	`np.percentile()`
6	Orrin	A whole range at once	2-D arrays, `axis=`, `[row, col]`
7 ⭐	The Queen	What if?	`np.where()`, `np.maximum()`, a function

You're done when...

- [ ] `my_work/` has `ex1` to `ex6` (and `ex7` if the Privy Council can wait), each printing the expected output
- [ ] Dalvric's quartile report runs and prints the expected three lines, with one commit per bug
- [ ] Everything is pushed to `lesson-08` and you've opened a pull request



Next lesson: The Living Ledger. pandas gives you a DataFrame: a whole table with named columns, filters, sorting and reading straight from Excel. The Treasury workbook, come to life.